The Art of Finance

Finance Overview · Time Value of Money · Debt · Equity · Alternatives · Applications

A structured learning journey

Master Finance from First Principles

A progressive curriculum that takes you from zero financial knowledge to confident investment analysis — built concept by concept, layer by layer.

Start Learning · View Curriculum

LEARNING PATH

01 Finance Overview
02 Time Value of Money
03 Debt
04 Equity
05 Alternatives
06 Applications

No experience needed · ~ Expert

First Principles — Every concept is grounded in fundamentals — no shortcuts, no jargon without explanation.

Progressive Depth — Each module builds on the last. You will never be asked to use a concept before it is introduced.

Visual Learning — Diagrams, formulas, and worked examples make abstract ideas concrete and memorable.

CURRICULUM

Your Learning Journey

Six modules that compose a complete picture of finance — each one a prerequisite for the next.

01 · Foundation
Finance Overview
What is finance? Understand markets, capital, and the role money plays in the economy.

02 · Core Concept
Time Value of Money
A dollar today is worth more than a dollar tomorrow. Learn present value, future value, and discounting.

03 · Instruments
Debt
Explore loans, bonds, and fixed-income instruments. Learn how to price debt using discounted cash flows.

04 · Instruments
Equity
Understand stocks, dividends, and ownership claims. Learn DCF valuation and relative pricing models.

05 · Advanced
Alternatives
Real assets, commodities, and alternative strategies. Portfolio theory and the risk/return tradeoff.

06 · Capstone
Applications
Put it all together. Evaluate real investments using the frameworks you have built from first principles.

Ready to begin?

No prior experience required. Start with an overview of what finance is and why it matters.

Begin: Finance Overview →

The Art of Finance
A first-principles approach to understanding finance, built layer by layer.

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Module 01 · Foundation

Finance Overview

Before diving into formulas and instruments, we must understand what finance is, why it exists, and the role it plays in the economy.

~ 2 sections · ~ 6 key terms · ~ No prerequisites

1 What is Finance?

At its core, **finance is the science of managing money across time.** It encompasses the decisions individuals, businesses, and governments make about acquiring and deploying capital to achieve their objectives.

Finance asks three fundamental questions:

Where does money come from?
How capital is raised — through debt, equity, or retained earnings.

Where does money go?
How capital is deployed — which projects, assets, or investments receive funding.

How is value created?
Whether the returns generated justify the risks and costs of the capital deployed.

THE FLOW OF CAPITAL

Savers & Investors → Financial Markets → Businesses & Governments → Returns & Growth

2 Three Domains of Finance

Personal Finance
Managing individual income, savings, investment, and debt. The decisions you make every day.
- Budgeting
- Retirement savings
- Mortgage
- Tax planning

Corporate Finance
How companies raise capital, allocate resources, and maximize shareholder value.
- Capital structure
- Mergers & acquisitions
- Dividend policy
- Valuation

Capital Markets
The markets where buyers and sellers trade financial instruments like stocks and bonds.
- Stock markets
- Bond markets
- Derivatives
- IPOs

3 Key Terms

These six terms will appear throughout every module. Click each to expand its definition.

Capital +
Risk +
Return +
Liquidity +
Valuation +
Yield +

You completed
Finance Overview

Next: Time Value of Money →

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Module 02 · Core Concept

Time Value of Money

The single most important concept in all of finance: a dollar received today is worth more than a dollar received in the future.

~ Prerequisite: Time Value of Money

1 Why Does Time Affect Value?

Money has time value for three reasons:

Opportunity Cost
Money available today can be invested to earn a return. Waiting sacrifices that return.

Inflation
Prices generally rise over time, eroding the purchasing power of future money.

Uncertainty
A payment promised in the future carries risk — it might not materialize. Certain cash now is worth more than uncertain cash later.

VALUE OF $1,000 OVER TIME AT 8% ANNUAL RETURN

2 The Core Formulas

Future Value

$$FV = PV \times (1 + r)^n$$

Given a present sum **PV**, a rate **r**, and **n** periods, how much will it be worth in the future?

PV = Present Value (today's amount)
r = Interest rate per period
n = Number of periods

Present Value

$$PV = FV / (1 + r)^n$$

Given a future cash flow **FV**, what is it worth in today's dollars? This is called discounting.

FV = Future Value (future amount)
r = Discount rate
n = Number of periods

Net Present Value (NPV)

Most investments produce a series of cash flows over time. NPV sums the present value of all future cash flows and subtracts the initial investment.

$$NPV = \sum [CF_t / (1 + r)^t] - C_0$$

If NPV > 0, the investment creates value. If NPV < 0, it destroys value.

3 Interactive Calculator

Explore how time, rate, and principal interact.

Compute Future Value · Compute Present Value

Present Value ($) 1000 · Annual Rate (%) 8 · Years (n) 5

Future Value **$1,469.33** · Rate: 8% / yr · Periods: 5 yrs · 1000 × (1 + 0.08)^5

← Finance Overview

Next: Debt →

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Module 03 · Instruments

Debt: Loans & Bonds

Debt is a promise to repay. Learn how loans work, how bonds are structured, and how to price fixed-income instruments using present value.

~ Prerequisite: Time Value of Money

1 What is Debt?

When an entity — a person, company, or government — needs money, one option is to **borrow it.** In exchange for cash today, they promise to repay the principal, typically with interest over time. This contract is called a **debt instrument.**

Debt is the lender's asset and the borrower's liability. The lender earns a return (interest). The borrower gets capital to deploy. Both parties benefit — if the interest rate is priced correctly for the risk.

ANATOMY OF A LOAN

Principal — The original amount borrowed.
Interest Rate — The cost of borrowing, per period.
Term — The duration of the loan.
Amortization — Schedule of principal repayment.

2 Understanding Bonds

A **bond** is a tradeable debt instrument. The issuer borrows from investors by selling bonds; investors receive periodic **coupon payments** (interest) and the **face value** (principal) at maturity.

Cash flows on a 3-year, 5% coupon, $1,000 face bond:

Type	Issuer	Risk	Typical Yield
Government Bond	National governments	Very Low	2–5%
Municipal Bond	State/local governments	Low	3–5%
Investment Grade Corp.	Large, stable companies	Moderate	4–7%
High Yield (Junk)	Speculative companies	High	7–15%

3 Bond Pricing Formula

A bond's price is simply the **present value of all its future cash flows** — the coupon payments plus the face value — discounted at the market yield (YTM).

$$P = \sum [C / (1+y)^t] + F / (1+y)^n$$

Where C = coupon payment, y = yield to maturity, F = face value, n = periods to maturity

The Price-Yield Relationship
When market interest rates **rise,** existing bond prices **fall** — and vice versa. This inverse relationship is one of the most important in fixed income.

BOND PRICING CALCULATOR

Face Value ($) 1000 · Coupon Rate (%) 5 · YTM (%) 6 · Years to Maturity 5

Bond Price **$957.88** · Annual Coupon $50.00 · Status: Trading at a Discount

← Time Value of Money

Next: Equity →

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Module 04 · Instruments

Equity & Stock Valuation

Equity represents ownership. Learn what stocks are, what drives their value, and how analysts price them using fundamental valuation models.

1 What is Equity?

A share of **equity** (or stock) is a fractional ownership claim on a company's assets and future earnings. Shareholders benefit when the company grows, but they are last in line to be paid in a liquidation — after debt holders.

Unlike debt, equity has no fixed repayment schedule. The return is entirely driven by the company's performance.

Dimension	Debt	Equity
Claim	Fixed interest + principal	Residual profits / assets
Priority	Senior — paid first	Junior — paid last
Upside	Capped at contracted rate	Unlimited
Downside	Limited if secured	Total loss possible
Voting rights	None	Yes (common stock)
Return driver	Interest rate	Earnings growth + sentiment

CAPITAL STRUCTURE STACK

2 Valuation Models

The fundamental principle of equity valuation is the same as bond pricing: the value of a stock equals the **present value of all future cash flows** it will generate for its owners.

DDM **Dividend Discount Model**

Values a stock as the present value of all future dividends, assuming constant growth rate g.

$$P = D_1 / (r - g)$$

D_1 — Next year's dividend · r — Required rate of return · g — Dividend growth rate (perpetual)

Best for mature, dividend-paying companies with stable growth

DCF **Discounted Cash Flow**

Values a company by discounting projected free cash flows plus a terminal value back to the present.

$$P = \sum [FCF_t / (1+r)^t] + TV$$

FCF_t — Free cash flow in period t · r — Discount rate (WACC) · TV — Terminal value at horizon

Most flexible model — applies to any cash-flow-generating business.

3 Valuation Calculator

Estimate intrinsic value using two common methods.

DIVIDEND DISCOUNT MODEL

Annual Dividend ($), $_0$: 2.50
Required Return (%): 10
Growth Rate (%): 4

Intrinsic Value (DDM) **$41.67**

P/E RELATIVE VALUATION

Apply a market multiple to earnings per share. Simple, but dependent on the multiple chosen.

Earnings Per Share ($): 5.00
P/E Multiple (×): 20

Implied Price (P/E) **$100**
$5.00 × 20×

← Debt

Next: Alternatives →

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Module 05 · Advanced

Alternative Investments

Beyond stocks and bonds lie a rich set of asset classes — real estate, commodities, private equity, and more. Explore their role in a diversified portfolio.

1 Why Alternatives?

Traditional portfolios of stocks and bonds are exposed to the same economic forces — recessions, interest rate shifts, and market sentiment. **Alternative investments** often have lower correlation to traditional assets, providing true diversification.

THE DIVERSIFICATION PRINCIPLE

$$\sigma^2_{portfolio} = \Sigma_i \Sigma_j w_i w_j \sigma_i \sigma_j \rho_{ij}$$

Portfolio variance depends on the **correlation (ρ)** between assets. When ρ < 1, combining assets reduces total risk without proportionally reducing expected return. This is the mathematical foundation of diversification.

Asset	Stocks	Bonds	Real Estate	Gold	Crypto
Stocks	1.00	−0.15	0.56	0.09	0.35
Bonds	−0.15	1.00	0.18	0.38	−0.05
Real Estate	0.56	0.18	1.00	0.19	0.05
Gold	0.09	0.38	0.19	1.00	0.15
Crypto	0.35	−0.05	0.05	0.15	1.00

Approximate historical correlations (illustrative). Values closer to 0 indicate better diversification.

2 Alternative Asset Classes

Real Estate — Moderate risk
Real Asset
Physical property — residential, commercial, industrial. Returns come from rent income and price appreciation.
Typical Return: 8–10% · Liquidity: Low

Commodities — High risk
Real Asset
Raw materials — oil, gold, agricultural products. Prices driven by supply/demand and macro conditions.
Typical Return: 3–8% · Liquidity: High

Private Equity — High risk
Private Markets
Investment in private companies. Higher return potential than public markets, but capital is locked up for years.
Typical Return: 12–20% · Liquidity: Very Low

Hedge Funds — Variable risk
Alternative Strategy
Pooled investment funds using sophisticated strategies — long/short, arbitrage, macro — to generate absolute returns.
Typical Return: Variable · Liquidity: Low

Cryptocurrencies — Very High risk
Digital Asset
Decentralized digital assets. High volatility, speculative in nature. Emerging asset class with evolving regulatory environment.
Typical Return: Highly Variable · Liquidity: High

3 Portfolio Allocation Explorer

Compare three common allocation strategies and their risk/return profiles.

Conservative · Balanced · Aggressive

50% Stocks · 35% Bonds · 15% Alts

Expected Return: 7.4% · Risk Level: Moderate · Alternatives: 15%

← Equity

Next: Applications →

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Module 06 · Capstone

Applications in Practice

Everything you have learned now comes together. Here we apply first-principles thinking to evaluate a real investment from scratch — the way a professional analyst would.

~ Prerequisite: All prior modules

What You Have Built

01 **Finance Overview** — Markets, capital flows, the role of finance
02 **Time Value of Money** — PV, FV, discounting, NPV
03 **Debt** — Bonds, yield, fixed-income pricing
04 **Equity** — Stocks, DDM, DCF, P/E valuation
05 **Alternatives** — Diversification, portfolio theory, all assets

1 A 5 Step Investment Framework

Professional investors follow a structured process. Click each step to explore it.

Understand the Business
Assess the Competitive Position
Analyze the Financials
4. Value the Investment
5. Make a Decision

1. Understand the Business
1. What does the company do — what product or service does it sell?
2. Who are its customers, and what is the addressable market?
3. What are the key drivers of revenue and cost?

Key Tools
Annual reports, 10-K filings, industry research

2 Worked Example: StableCo Inc.

A fictional mature consumer staples company with steady cash flows and a 25-year dividend history.

COMPANY DATA

Annual Dividend $3.20 · Dividend Growth Rate 4% / yr · Required Return 9% · Current P/E 18× · EPS (ttm) $6.40 · Net Debt / EBITDA 1.8×

DDM Valuation $64.00
P = $3.20 × (1.04) / (9% − 4%)

P/E Valuation $6.40 × 18× $115.20

VALUATION SUMMARY

Average Intrinsic Value **$89.60** · Current Market Price **$100.00** · Margin of Safety **-22.8%**

At $100.00, StableCo appears to be trading above our estimated intrinsic value. There is no margin of safety at this price — a buyer is either paying a premium for quality, or the assumptions need revisiting.

Journey Complete

You have gone from first principles — what money is and why time affects its value — all the way to evaluating a real investment like a professional analyst. Finance is a craft, and you now have the foundation to keep building.

Back to Home · Review from Start

← Alternatives

Home →

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